

► **Intellectual Property:** Go to a beat down shop in one of your favourite bazaars and you'll find a bunch of replicas of your absolute favourite memorabilia – from Harry Potter key chains to V for Vendetta based Guy Fawkes' masks. But the availability of these items in sheer number, and the quality (or the lack of it) they offer do not really pose a great threat to the sales revenue of either Warner Bros. or Disney here, as you can argue. But then, 3D printers – having the capacity to create high quality imitations in every household – could well be the absolute ultimate threat to intellectual property. It is easy to translate absolutely any existing object into a digital file using a 3D scanner. Therefore, designer products could be sold at extremely low prices by just recreating them through 3D printers. This environment would make it very difficult for businesses to be able to fully monetize their inventions, or for media giants – their creations. IP thieves will have lesser costs, sell counterfeited goods at discounts, and sell poorly performing or even dangerous products, to unsuspecting customers. According to forecasts, about \$100 billion could be lost in IP theft per year thanks to 3D printers.

So, as is evident, there are many ways of misusing this potentially world-changing technology, but it is very important for the various governments, while contemplating the dangers of this era, to avoid overreacting in ways they end up hindering the process of innovation and unleashing unintended consequences. Perhaps methods those are smart and not necessarily radical can be implemented to curb the misuse of this technology as mentioned above. For example, gun laws could now become gunpowder laws – meaning there could be laws to monitor the usage of gunpowder. This could be a better way of controlling the usage of firearms rather than trying to keep a tab on the endless permutations of gun designs. A response to the key hacker problem could be to make advances in biometric scanning, or other cheaper, more advanced forms of security – a very beneficial evolution on its own.

In short, the consumers and the manufacturers are both going to go through a revolution of sorts with the widespread introduction of 3D printers. And as with any revolutionary technology, there are endless possibilities – both good and bad – that 3D printing can bring in the future. The challenges that 3D printing poses can be dealt with only by increased creativity and flexibility, but it can't be denied that the inevitable dawn of this new era brings more scope for positive growth than otherwise. 